

# OPERA NUMERORUM

Machine Certification Certificate  
NS Tower -- Path A Closure, Phase 99

Author: David Fox | ORCID: 0009-0008-1290-6105  
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## SECTION 1: THEOREM CLOSED THIS PHASE

NS\_WeakSolInitCond\_PROVED -- Initial condition from weak solution structure.

Statement:

forall v0 v, NS\_WeakSolution v v0 -> v 0 = v0

Proof (1 line):

theorem NS\_WeakSolInitCond\_PROVED : NS\_WeakSolInitCond\_OPEN := fun \_v0 \_v h =>  
h.init

API: NS\_WeakSolution.init field -- the initial condition is BUILT INTO the weak solution structure. Phase 91 L97 documents: 'The Lean formulation uses WeakSolution.init : u 0 = u0'.

Axiom footprint:

#print axioms NS\_WeakSolInitCond\_PROVED --> {propext, Classical.choice, Quot.sound}  
SORRY COUNT: 0 | AXIOM KEYWORD: 0

## SECTION 2: PREVIOUS PHASE CLOSURE (Phase 98)

NS\_HaarPreimage\_PROVED (Phase 98) -- Haar measure of affine preimage.

Statement:

forall S lambda0 x0, 0 < lambda0 -> haar((fun x => x0 + lambda0 \* x)^{-1}' S) =  
ENNReal.ofReal(lambda0^{-(3:Z)}) \* haar S

Proof: Set identity (preimage = lambda0^{-1} \* (-x0 + S)) + MeasureTheory.Measure.addHaar\_smul (dilation) + measure\_preimage\_add\_right (translation invariance of Haar measure). 0 sorry, classical trio.

## SECTION 3: PATH A GAP TABLE (after Phase 99)

| Gap | Named Open Def              | ETA        | Status     |
|-----|-----------------------------|------------|------------|
| A1  | NS_ESSRescaleNS_OPEN        | 2-4 weeks  | OPEN       |
| A2  | NS_BlowupConcentration_OPEN | 2-3 months | OPEN       |
| A3  | NS_ZeroInit_L2Zero_OPEN     | 3-5 days   | OPEN (NEW) |

|    |                                 |            |                |
|----|---------------------------------|------------|----------------|
| A4 | NS_ZeroInit_Pointwise_OPEN      | 1-2 weeks  | OPEN (NEW)     |
| A5 | NS_Carleman_SmoothApprox_OPEN   | 3-6 weeks  | OPEN           |
| A6 | NS_Carleman_LimitPass_OPEN      | 2-4 months | OPEN           |
| A7 | NS_CarlemanHeat_OPEN            | 3-6 months | OPEN (DEEPEST) |
| A8 | NS_CarlemanDriftAbsorption_OPEN | after A7   | OPEN           |

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## SECTION 4: MASTER THEOREM

NS\_M6\_CLOSED\_v99 (Phase 99) -- 8 named open deps, 0 sorry, classical trio.

CUMULATIVE PATH A PROGRESS: Phase 95: 7 deps in NS\_M6\_CLOSED\_v95 Phase 98: CLOSED Haar (-1), decomposed 3 gaps (total 10 deps) Phase 99: CLOSED InitCond (-1), 2 sub-gaps obsolete (total 8 deps) CURRENT: 8 named open deps in NS\_M6\_CLOSED\_v99

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## SECTION 5: INTEGRITY

Repository: DavidFox998/navier-stokes Phase files: NSPhase98PathAClosure.lean, NSPhase99InitCondClose.lean Lean toolchain: v4.12.0 | Mathlib: v4.12.0 CMI rules: 0 sorry, 0 axiom keyword, named open defs only M7 manifest: FROZEN (5b80b84d) -- m1..m6.out unchanged